CLINICAL · APPROACH TO

Approach to nasal obstruction

Nasal obstruction is the commonest nasal complaint, affecting around a third of adults, and its causes fall into two families. **Inflammatory** obstruction from rhinitis and chronic rhinosinusitis fluctuates, alternates and is usually bilateral; **structural** obstruction from the septum and the nasal valves is fixed and often one-sided. Most patients have one or both, and the approach is to tell them apart, treat the mucosa before operating on the skeleton, and never miss the unilateral blockage with blood or a foul discharge that is a tumour or, in a child, a foreign body.

AT A GLANCE

- Fixed and one-sided: structural or a mass.
- Fluctuating and two-sided: mucosal.
- Children: adenoids, foreign body, choanal atresia.
- Unilateral blockage with bleeding or pain: a tumour until proved otherwise.

1 CHAPTER ONE How the nose blocks

Resistance, valves and mucosa.

1.1 The physiology that matters

The nose provides about two-thirds of total airway resistance, and most of it lies at the **internal nasal valve**, the slit between the septum, the caudal edge of the upper lateral cartilage and the head of the inferior turbinate. It is the narrowest part of the airway, and because resistance rises steeply as the radius falls (Poiseuille's law), a small narrowing here causes a large symptom. The valve is also richly supplied with receptors, so obstruction at this point feels worse than obstruction further back.

The mucosa of the turbinates is erectile tissue under sympathetic control. In about 80% of people it congests and decongests alternately on each side in the **nasal cycle**, every two to six hours, keeping total resistance constant. A patient who notices the cycle, or whose fixed deviation is unmasked by the congested phase, describes alternating blockage.

The sensation of airflow is carried by trigeminal cold receptors, not by the volume of air. This is why the subjective complaint and objective measures often disagree, why menthol makes the nose feel clearer, and why a nose that has lost too much turbinate can feel blocked although it is wide open (*empty nose syndrome*).

DO NOT MISS

- **Unilateral obstruction with blood-stained discharge, pain, facial swelling, numbness, loose teeth, diplopia or a neck node:** sinonasal or nasopharyngeal malignancy.
- **An adolescent boy with unilateral obstruction and heavy epistaxis:** juvenile angiofibroma — do not biopsy in clinic.
- **A child with unilateral foul discharge:** foreign body.
- **A newborn with cyclical cyanosis relieved by crying:** bilateral choanal atresia.
- **A midline nasal mass in a child:** dermoid, glioma or encephalocele — image before any biopsy.

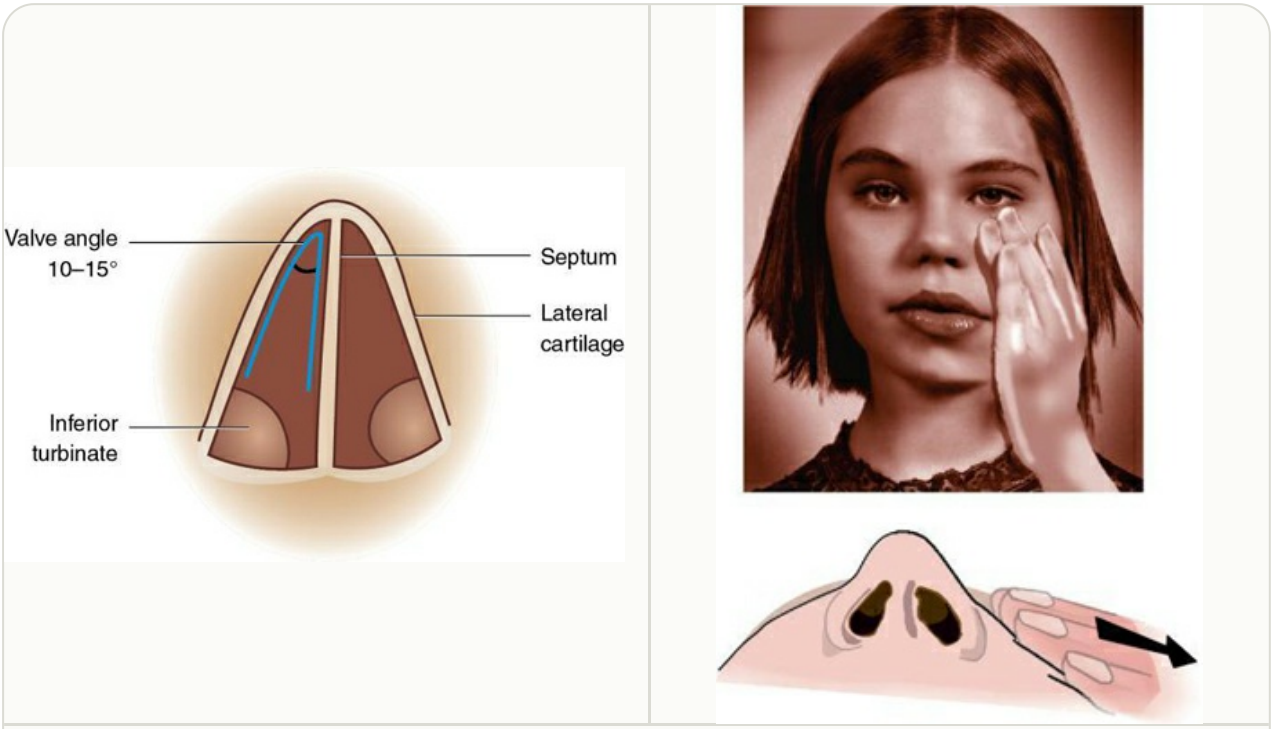


FIGURE 1 First, the internal nasal valve area. Second, Cottle's test: drawing the cheek laterally opens the valve.
Dhingra, figs 23.3 and 26.7

2 CHAPTER TWO History

Fixed or fluctuating?

2.1 The questions

Ask whether the blockage is one-sided or both, constant or variable, and whether it alternates. **Fixed, unilateral** obstruction points to the septum, a valve problem, a polyp, a tumour or a foreign body; **fluctuating, bilateral** obstruction points to rhinitis or chronic rhinosinusitis. Ask about onset (after trauma or surgery, or gradual), and whether the patient sniffs inward and the nostril collapses, which suggests valve collapse.

Ask about the company the symptom keeps: sneezing, itch and watery discharge in rhinitis; loss of smell, facial pressure and discoloured discharge in rhinosinusitis; bleeding, pain and crusting in tumours, granulomatous disease and septal perforation; and snoring and mouth breathing. Take a full drug history, including decongestant sprays (rhinitis medicamentosa), antihypertensives, the oral contraceptive pill and recreational cocaine, and ask about asthma, aspirin sensitivity, smoking and occupation. The **NOSE** score and **SNOT-22** quantify the symptom and its effect on life.

3 CHAPTER THREE Examination

Outside, then inside, then after decongestion.

3.1 The nose

Look at the external nose from the front, side and below for deviation, a twisted pyramid, a narrow or collapsing ala and a caudal septum dislocated into the nostril. Watch the alae on deep nasal inspiration. **Cottle's test**, in which the cheek is drawn laterally to open the valve, suggests a valve problem when it improves breathing; a modified version supports the valve from inside with a cotton bud or a curette.

Examine the vestibule, the septum and the turbinates with a headlight and speculum, then with an endoscope before and after topical decongestant. Mucosal swelling that shrinks well is reversible (rhinitis); a deviation, a spur or a polyp does not change. The endoscope also reveals middle meatal pus and polyps, the posterior septum, the choanae and the nasopharynx.

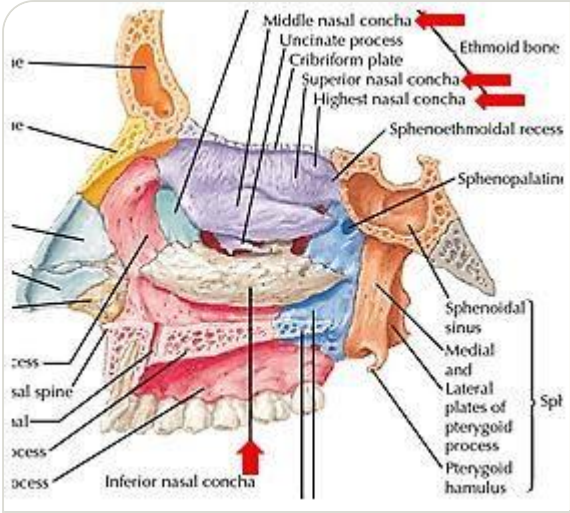


FIGURE 2 The lateral nasal wall: turbinates and meatuses.
Anatomy atlas · ENT Steps p. 99

3.2 Measuring the airway

Peak nasal inspiratory flow is the quickest objective test, normally around 120 to 140 L/min in adults. **Anterior rhinomanometry** measures resistance from flow and pressure and is the reference standard. **Acoustic rhinometry** maps cross-sectional area along the cavity; a large improvement after decongestion points to mucosal disease. CT is reserved for suspected sinus disease or before surgery, and imaging with MRI for any suspected tumour.

4 CHAPTER FOUR Causes

Inflammatory, structural, and the rest.

CAUSES OF NASAL OBSTRUCTION

GROUP	CAUSES	TYPICAL CLUE
Inflammatory	Allergic and non-allergic rhinitis; rhinitis medicamentosa; chronic rhinosinusitis with or without polyps	Bilateral, fluctuating, shrinks with decongestant
Septum	Deviation, spur, dislocated caudal septum, haematoma	Fixed, unilateral or S-shaped
Valves	Internal valve narrowing (high septal deviation, weak upper laterals, post-rhinoplasty); external valve collapse	Worse on inspiration; Cottle positive
Turbinates	Hypertrophy, concha bullosa	Seen on endoscopy, partly reversible
Masses	Polyps, antrochoanal polyp, inverted papilloma, malignancy, angiofibroma	Unilateral, progressive; blood
Granulomatous	Granulomatosis with polyangiitis, sarcoidosis, cocaine	Crusting, bleeding, perforation
Other	Adenoids, choanal atresia, foreign body, empty nose syndrome, pregnancy and hypothyroidism	Age and context

4.1 Rhinitis and rhinosinusitis

Allergic rhinitis affects about a quarter of people and chronic rhinosinusitis about one in ten. Allergic rhinitis mainly swells the inferior turbinates and responds well to decongestant; chronic rhinosinusitis affects the middle meatus and responds less. Both are common partners of a deviated septum, and treating the mucosa often settles symptoms without surgery.

4.2 The septum and the valves

Cottle's classification divides the septum into five areas from the nostril backwards. Deviations in areas 1 to 3, at or in front of the internal valve, cause the most obstruction; deviations further back matter less. A deviation at the valve or the caudal septum is not adequately corrected by a standard septoplasty and needs a **functional septorhinoplasty**, with spreader grafts for the internal valve and struts or alar grafts for the external valve.

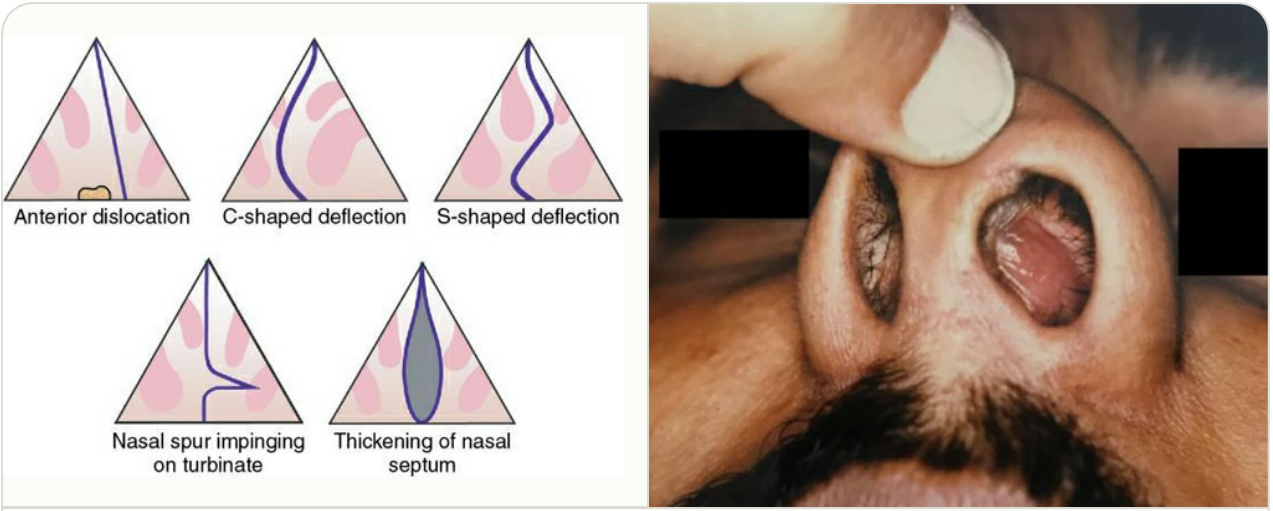


FIGURE 3 First, the types of septal deviation. Second, a C-shaped deviation nearly blocking one side.

Dhingra, figs 26.4 and 26.6

5 CHAPTER FIVE Pathway

Mucosa or skeleton?

THE BLOCKED NOSE

1 Red flags?

→ **unilateral with blood, pain or cranial nerve signs**
endoscopy, MRI and CT, biopsy unless angiofibroma suspected

→ **adolescent boy with bleeding**
contrast imaging first, no biopsy

→ **child with unilateral foul discharge**
remove the foreign body

2 Fixed or fluctuating?

→ **bilateral, fluctuating, shrinks with decongestant**
rhinitis — treat the mucosa

→ **smell loss, discharge, polyps**
chronic rhinosinusitis pathway

→ **fixed or unilateral**
step 3

3 Where is the narrowing?

→ **mid or posterior septum**
septoplasty

→ **high or caudal septum, collapsing valve**
functional septorhinoplasty

→ **turbinates**
medical first, then reduction

4 Still blocked after surgery or with a wide nose?

→ **recurrent deviation or valve collapse**
revision

→ **wide cavity with paradoxical obstruction**
empty nose syndrome — avoid further resection

CHAPTER SIX

6 Management

Treat the mucosa first.

6.1 Medical treatment

Every patient with an inflammatory component should have a proper trial of saline irrigation and a topical steroid spray, used correctly with the head bent forward and

the nozzle aimed away from the septum, for at least six to eight weeks. Antihistamines, allergen avoidance, a leukotriene antagonist in selected patients, and immunotherapy for confirmed allergy follow. Decongestant sprays are limited to a few days, because rebound congestion (*rhinitis medicamentosa*) follows longer use. Chronic rhinosinusitis is treated on its own pathway.

6.2 Surgery

A randomised trial has shown that septoplasty, with or without turbinate reduction, improves nasal obstruction more than continued medical treatment in patients with a deviated septum. **Septoplasty** corrects mid and posterior deviations through mucoperichondrial flaps, preserving an L-shaped strut of at least 1 cm dorsally and caudally. **Turbinate reduction** by submucosal diathermy, radiofrequency, outfracture or submucosal resection reduces the turbinates while preserving mucosa; its effect fades with time and it does not replace medical treatment. **Functional septorhinoplasty** treats the valve and the twisted nose, often through an external approach. Over-resection of the turbinates is avoided because of the risk of empty nose syndrome.

6.3 Children

In a child, adenoidal hypertrophy is the commonest cause of bilateral obstruction, with mouth breathing, snoring and hyponasal speech. Rhinitis is common, and a unilateral foul discharge is a foreign body until proven otherwise. A neonate who is an obligate nasal breather and blocked may have **choanal atresia**: bilateral atresia causes cyclical cyanosis relieved by crying and needs an oral airway, CT and endoscopic repair, while unilateral atresia presents later with a unilateral discharge. **Piriform aperture stenosis**, nasolacrimal duct cysts and midline masses (dermoid, glioma, encephalocele) are rarer neonatal causes that need imaging before intervention.

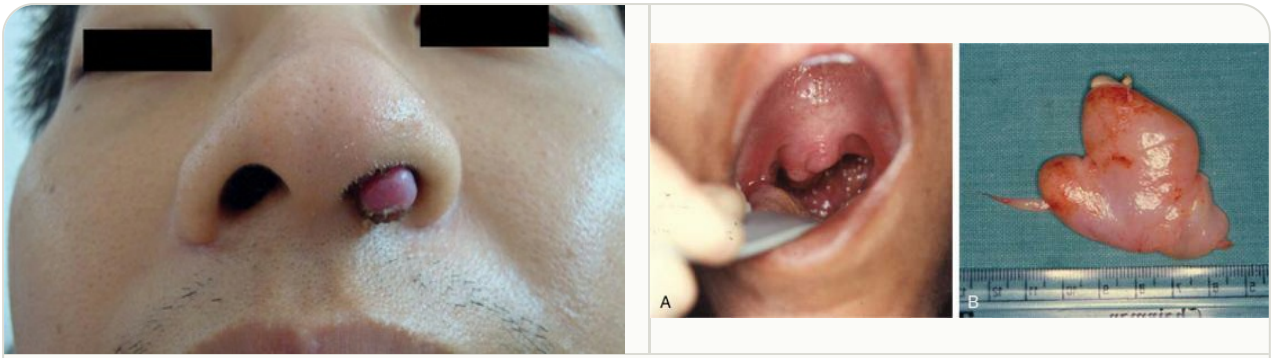


FIGURE 4 An antrochoanal polyp, a cause of unilateral obstruction in older children and young adults.

Dhingra, figs 32.1 and 32.2

7 CHAPTER SEVEN For the examination

What the examiner wants to hear.

7.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **The internal nasal valve:** boundaries and why it matters.
- **Inflammatory versus structural obstruction:** the decongestant test.
- **Cottle's test and Cottle's areas.**
- **Objective tests:** peak nasal inspiratory flow, rhinomanometry, acoustic rhinometry.
- **Septoplasty versus functional septorhinoplasty;** the L-strut.
- **Red flags** for unilateral obstruction; choanal atresia in the newborn.

7.2 Common pitfalls

COMMON PITFALLS

- Operating on a deviated septum when the obstruction is rhinitis that was never treated.
- Performing a standard septoplasty for a high or caudal deviation that needs valve surgery.

- Biopsying a vascular mass in an adolescent boy in clinic.
- Over-reducing the turbinates.

Written for this app from Oxford Textbook of Otolaryngology ch. 33, 78, Oxford Specialist Handbook ch. 3, Oxford Handbook of ENT ch. 3, Dhingra ch. 26, 28. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

IMAGING

On imaging

The imaging of this topic: key findings, differentials and representative scans.

NOSE & NASAL CAVITY · INFECTION & INFLAMMATION

Nasal Polyps

IMAGING FINDINGS

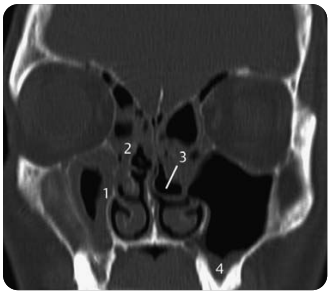
- Polypoid soft tissue filling the nose and all sinuses, both sides
- Widening of the ethmoid labyrinth and thinned septa
- Hyperdense contents suggest fungus or inspissated mucus

TOP DIFFERENTIALS

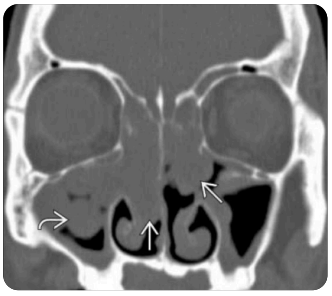
- Allergic fungal sinusitis
- Cystic fibrosis
- Inverted papilloma (unilateral)

CHECKLIST

- Bone erosion or asymmetry: think tumour or fungus
- Pre-operative CLOSE checklist



CT, coronal. The scan shows mucosal thickening in the right maxillary sinus (1), partial opacification of the ethmoid cells (2) on both sides, pneumatizati…



Coronal bone CT shows the classic appearance of sinonasal polyposis

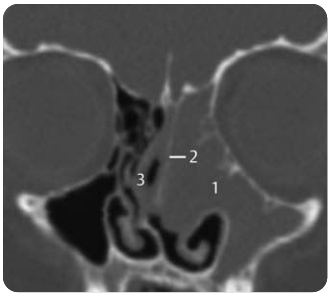
Koch et al., Diagnostic Imaging: Head and Neck, 3rd ed., p. 708

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 708.

NOSE & NASAL CAVITY · INFECTION & INFLAMMATION

Unilateral Nasal Polyps

- Destruction of bony outlines (with infiltrative characteristics) is suggestive of inverted papilloma, malignancies, or fungal infection.
- Calcifications in the opacified area are suggestive of fungal infection. Resorption of bone around tooth roots suggests odontogenic infection.
- Beware of pathology in the sphenoethmoidal fossa (i.e., juvenile angiofibroma).



CT, coronal. Complete opacification of the left maxillary sinus and ethmoid (1) is seen. Medial displacement of the middle turbinate (2) and nasal septu…

NOSE & NASAL CAVITY · TUMOURS

Inverted Papilloma

IMAGING FINDINGS

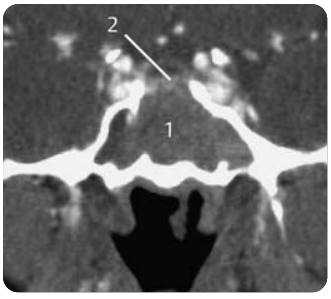
- Unilateral mass on the lateral nasal wall near the middle meatus
- Focal hyperostosis at the attachment
- Convoluted cerebriform pattern on T2 and post-contrast MRI

TOP DIFFERENTIALS

- Antrochoanal or solitary polyp
- Squamous cell carcinoma (in up to about 10%)
- Juvenile angiofibroma

CHECKLIST

- MRI separates tumour from secretions
- Bone destruction raises concern for carcinoma



Patient operated before because of nasal polyps. Histologic evaluation revealed inverted papilloma. Follow-up scans are shown.…



Axial bone CT shows a mass in the nasal cavity along the lateral wall near the middle meatus. The maxillary sinus is opacified, but it is difficult to…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 744.

NOSE & NASAL CAVITY · TUMOURS

Angiofibroma, Juvenile

IMAGING FINDINGS

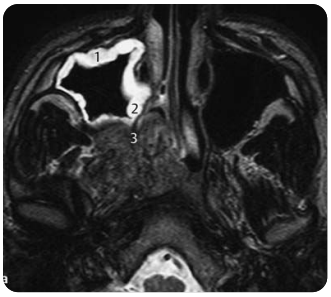
- Enhancing mass at the sphenopalatine foramen in an adolescent male
- Widened pterygopalatine fossa; bowing of the posterior maxillary wall
- Flow voids; extension to the sphenoid, infratemporal fossa and skull base

TOP DIFFERENTIALS

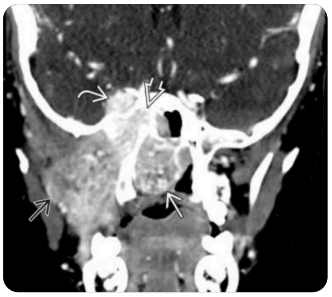
- Hypervascular polyp
- Rhabdomyosarcoma
- Antrochoanal polyp

CHECKLIST

- No biopsy: angiography and embolisation before surgery
- Vidian canal and cavernous sinus spread



A young man with obstruction and bloodtinged nasal discharge from the right side of the nose.
Dunnebier, Imaging for…



Coronal CECT in a 13-year-old shows typical growth pattern of a large, hypervascular angiofibroma occluding the nasopharynx, extending into righ…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 740.

NOSE & NASAL CAVITY · CONGENITAL

Choanal Atresia

IMAGING FINDINGS

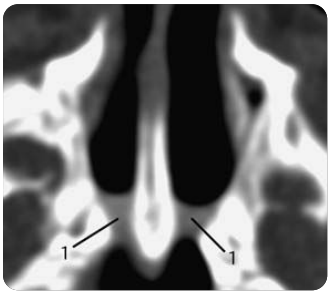
- Narrowed posterior choanae
- Thickened vomer; medially bowed posterior maxilla
- Bony, membranous or mixed closure; fluid in the nose

TOP DIFFERENTIALS

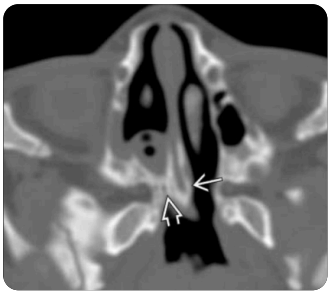
- Choanal stenosis
- Pyriform aperture stenosis
- Nasolacrimal duct mucocele

CHECKLIST

- CT after suction and decongestion
- CHARGE: look at the inner ears



CT, axial. Both choanal areas demonstrate narrowing of the bony passage, but congenital fibrous closure (1) is the cause of the distres…



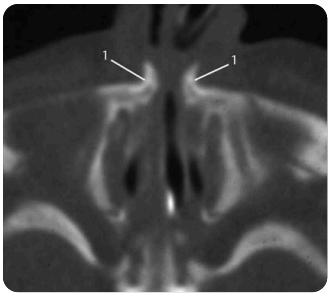
Axial bone CT through the upper choana in a child with mixed bony/osseous choanal atresia shows osseous choanal obstruction…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 670.

NOSE & NASAL CAVITY · CONGENITAL

Piriform Aperture Malformation

- A developmental disorder of this kind may result in concomitant pathologies, such as the fusion of both maxillary central incisors to form one megaincisor (Fig. 7.15), in the majority of patients. Holoprosencephaly and abnormalities of the pituitary–adrenal axis are also encountered.



CT, axial. The severe dyspnea was due to gross narrowing of the anterior nasal passage caused by bilateral bony stenosis of the piriform aperture (1)....